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103963

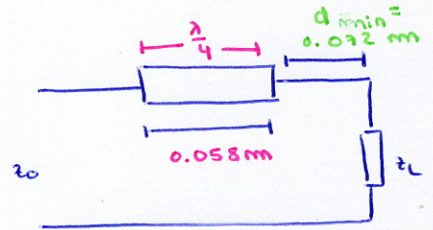
7. transformador $\frac{\lambda}{4}$ num ponto de mínimo

$$Z_L = 71.1 + 51.1j \quad (2)$$

$$Z_L = 1.42 + 1.022j \quad (2)$$

The Complete Smith Chart

Black Magic Design



$$\lambda = \frac{c}{f \sqrt{\epsilon_r}}$$

$$= \frac{3 \times 10^8}{962 \times 10^6 \times \sqrt{1.7779}}$$

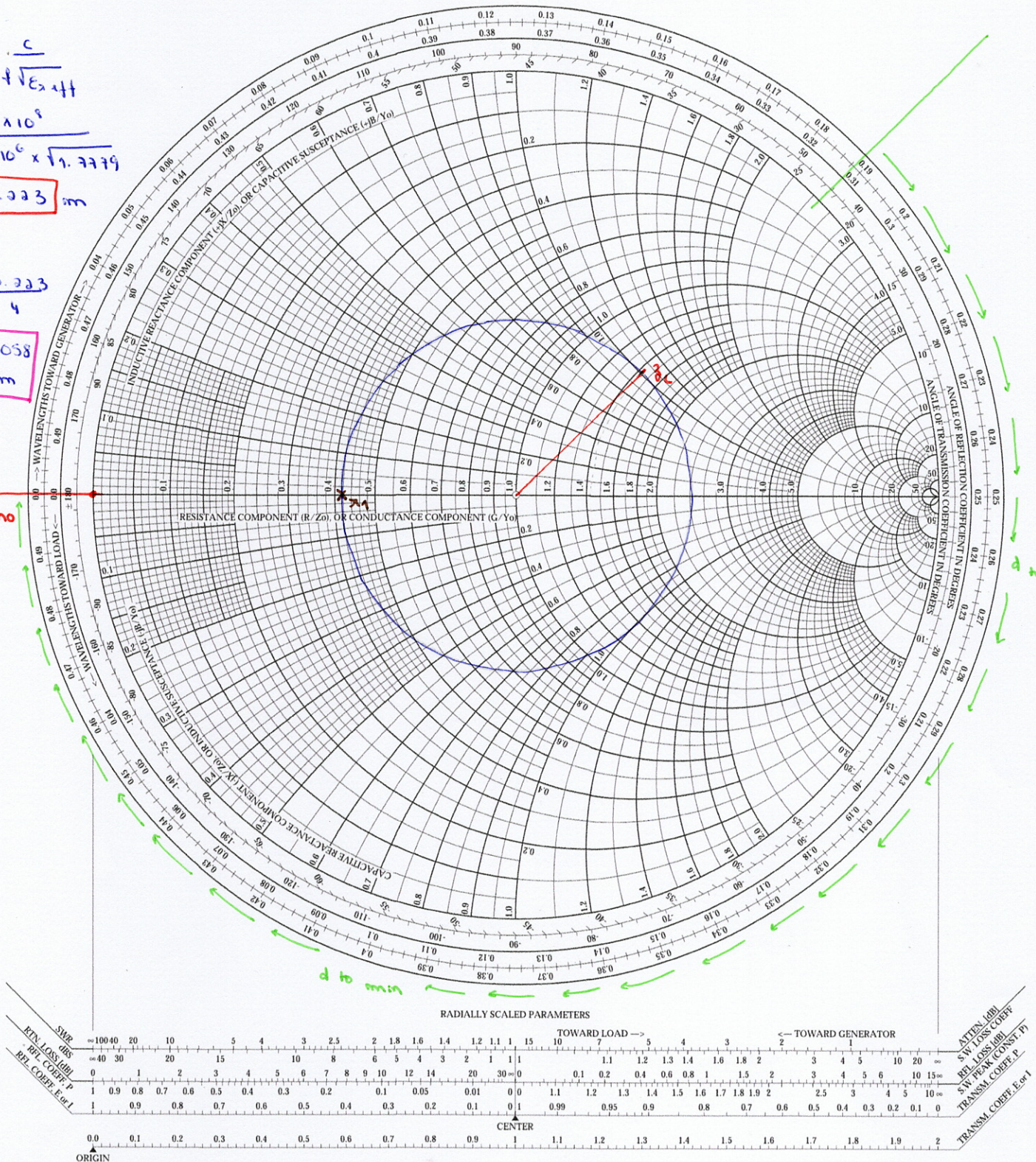
$$\lambda = 0.223 \text{ m}$$

$$\frac{\lambda}{4} = 0.0558 \text{ m}$$

$$\frac{\lambda}{4} = 0.058 \text{ m}$$

minimo

d to min



$$\eta_1 \Rightarrow \text{ponto mínimo} = 0.49\lambda$$

$$R_1 = \eta_1 \times 50 = 0.49 \times 50 = 20.5\lambda$$

$$R_1 = Z_0 = 20.5\lambda$$

$$d \text{ to min} = (0.5 - 0.189) \lambda$$

$$= 0.311 \lambda$$

$$d \text{ to min} = 0.072 \text{ m}$$

$$Z_1 = \sqrt{Z_0 Z_L}$$

$$= \sqrt{20.5 \times 50}$$

$$Z_1 = 32.07\lambda$$